

Healthcare Appointment No-Show Analysis

Project Overview

This project provides a comprehensive analysis of healthcare appointment attendance data to identify factors associated with patient no-shows. Missed appointments can lead to wasted clinical resources, longer waiting times, and reduced service efficiency. Using Excel and Power BI, the data was cleaned, analysed, and visualised to identify patterns and provide recommendations for enhancing patient attendance rates.

Data

This analysis was conducted using a dataset containing 110,527 healthcare appointment records obtained from kaggle.

Data Description

The dataset consists of the following columns:

- Gender
- Scheduled Day
- Appointment Day
- Age
- Clean Neighbourhood
- Scholarship
- Hypertension
- Diabetes
- Alcoholism
- SMS received
- No-show

Tools Used

- Microsoft Excel
- Power BI
- DAX (Data Analysis Expressions)
- Pivot Tables
- Data Cleaning and Transformation Techniques

Methodology

The data was extracted from Kaggle. Before analysing the data, I performed several cleaning and transformation steps. The dataset was first cleaned and prepared in Excel to ensure consistency and accuracy. Missing values and duplicate records were reviewed, and additional columns were created to support analysis, including Age Group, Attendance Status and SMS Status categories.

Pivot tables were used to calculate attendance and no-show rates across various patient demographics. The cleaned dataset was then imported into Power BI, where DAX measures were created to calculate key performance indicators such as the overall-no show rate.

Interactive visualisations were developed to analyse attendance patterns by age group, gender, SMS reminder status, and appointment day.

Key Findings

The following key insights were observed:

- Overall no-show rate: 20.19%
- Patients aged 18-35 recorded the highest no-show rate (23.82%).
- Patients aged 60+ recorded the lowest no-show rate (15.21%).
- Gender had minimal impact on attendance rates.
- Patients who received SMS reminders recorded a higher no-show rate than those who did not receive SMS reminders.
- Appointment day showed only a small impact on attendance behaviour.

Recommendations

Based on the insights derived from the data analysis, the following recommendations are made:

- Introduce appointment confirmation and rescheduling options.
- Investigate factors affecting SMS reminder effectiveness.
- Explore predictive analysis to identify high-risk appointments.
- Develop targeted interventions for younger patients.
- Continue monitoring attendance trends using dashboards.

Conclusion

This analysis identified age as the strongest factor associated with patient attendance, with younger patients demonstrating a higher chance of missing appointments. Gender and appointment day showed relatively little impact on attendance rates. The findings highlight opportunities for targeted interventions and demonstrate how data can be used to support service improvement and operational decision-making.